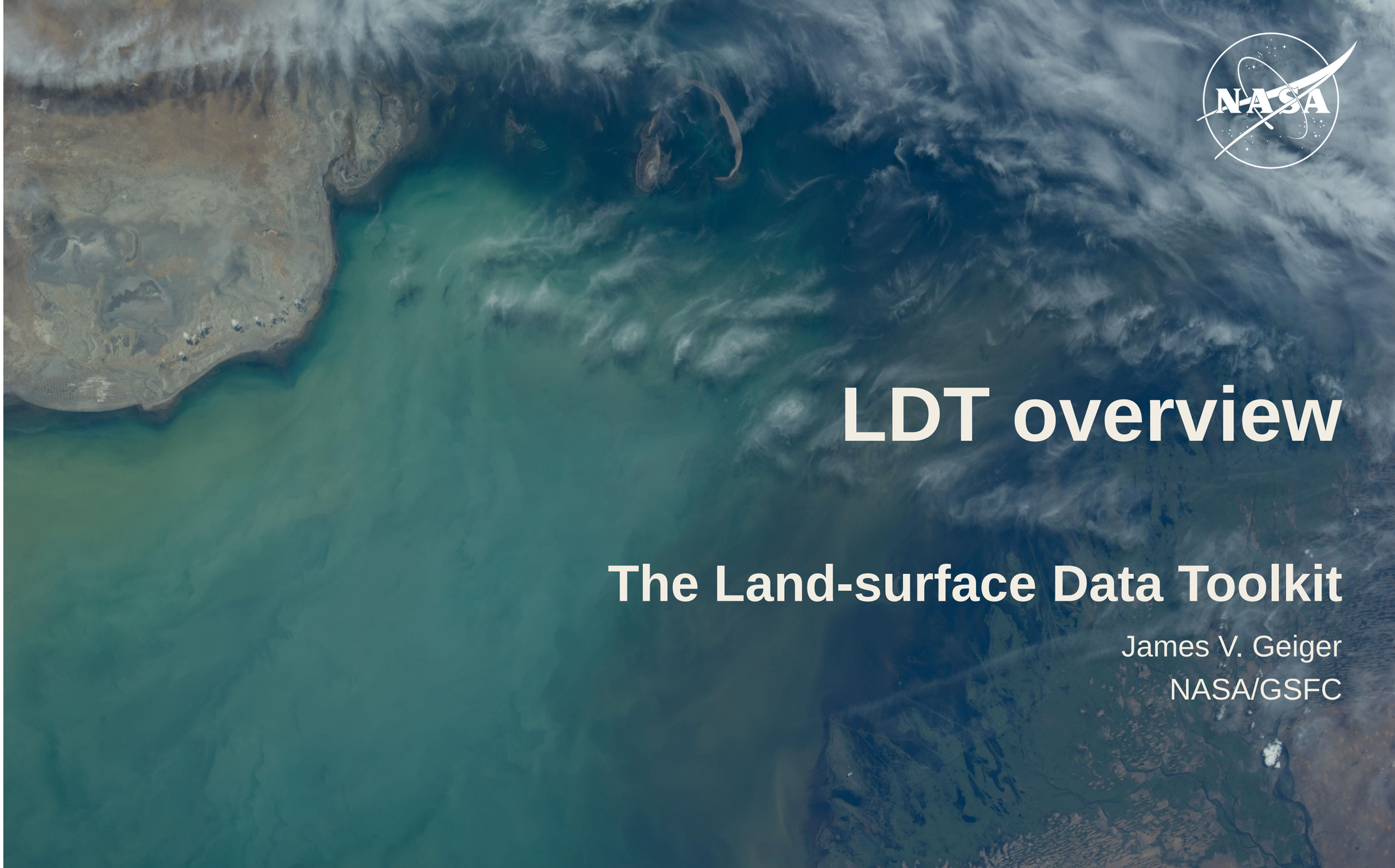


LDT overview

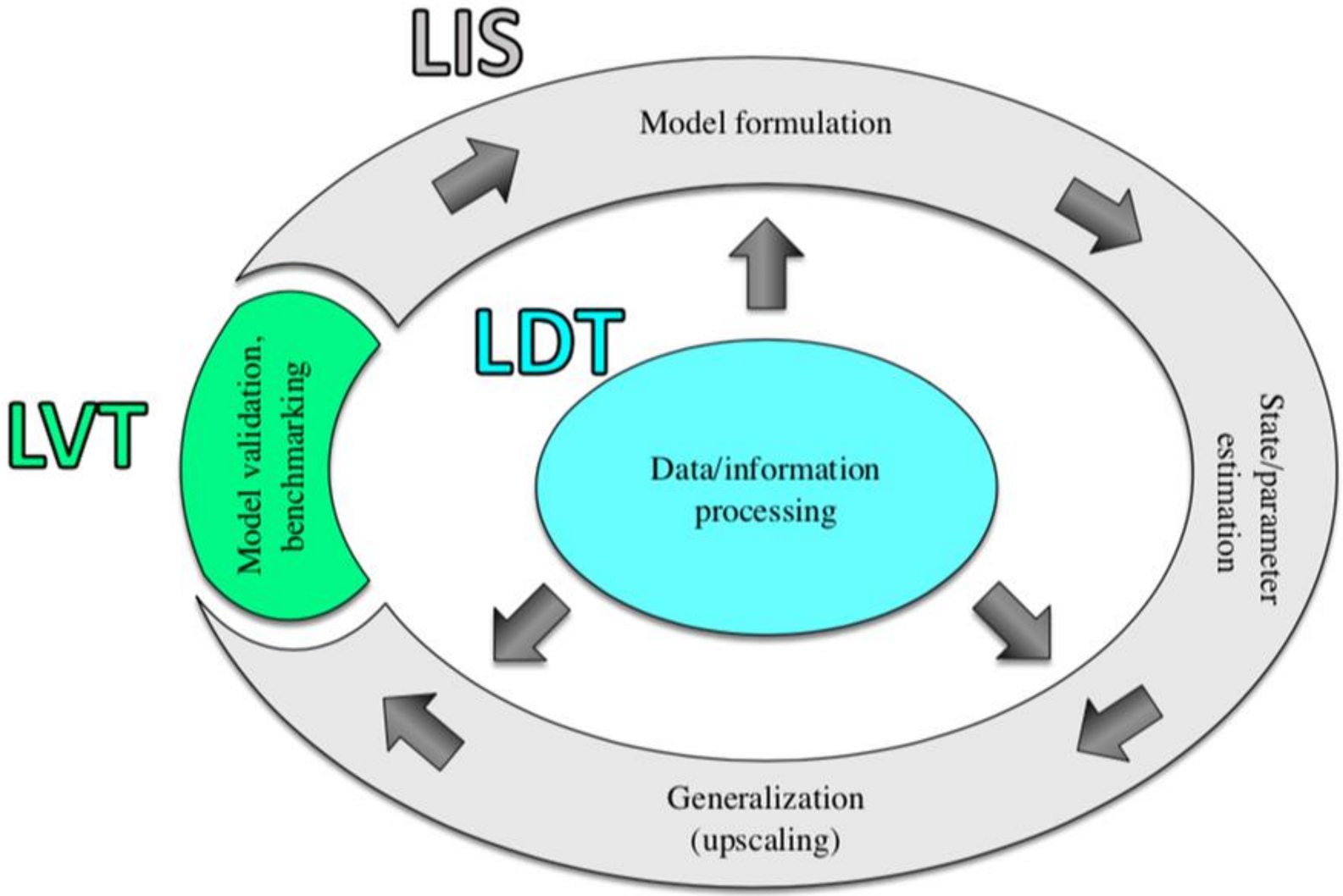
The Land-surface Data Toolkit

James V. Geiger
NASA/GSFC

The Land-surface Data Toolkit (LDT)

- The main preprocessing toolkit for LIS (and also LVT)
- Used to process groups of parameters needed for land surface and hydrological models
- A variety of input processing options
- Observation-based data assimilation options (e.g., CDF-matching)
- Generate ensemble or climatology-based initial conditions for different models
- Supports a variety of meteorological forcing datasets
- Temporal disaggregation of daily (or greater) precipitation fields
- Artificial Neural Network (ANN) capabilities

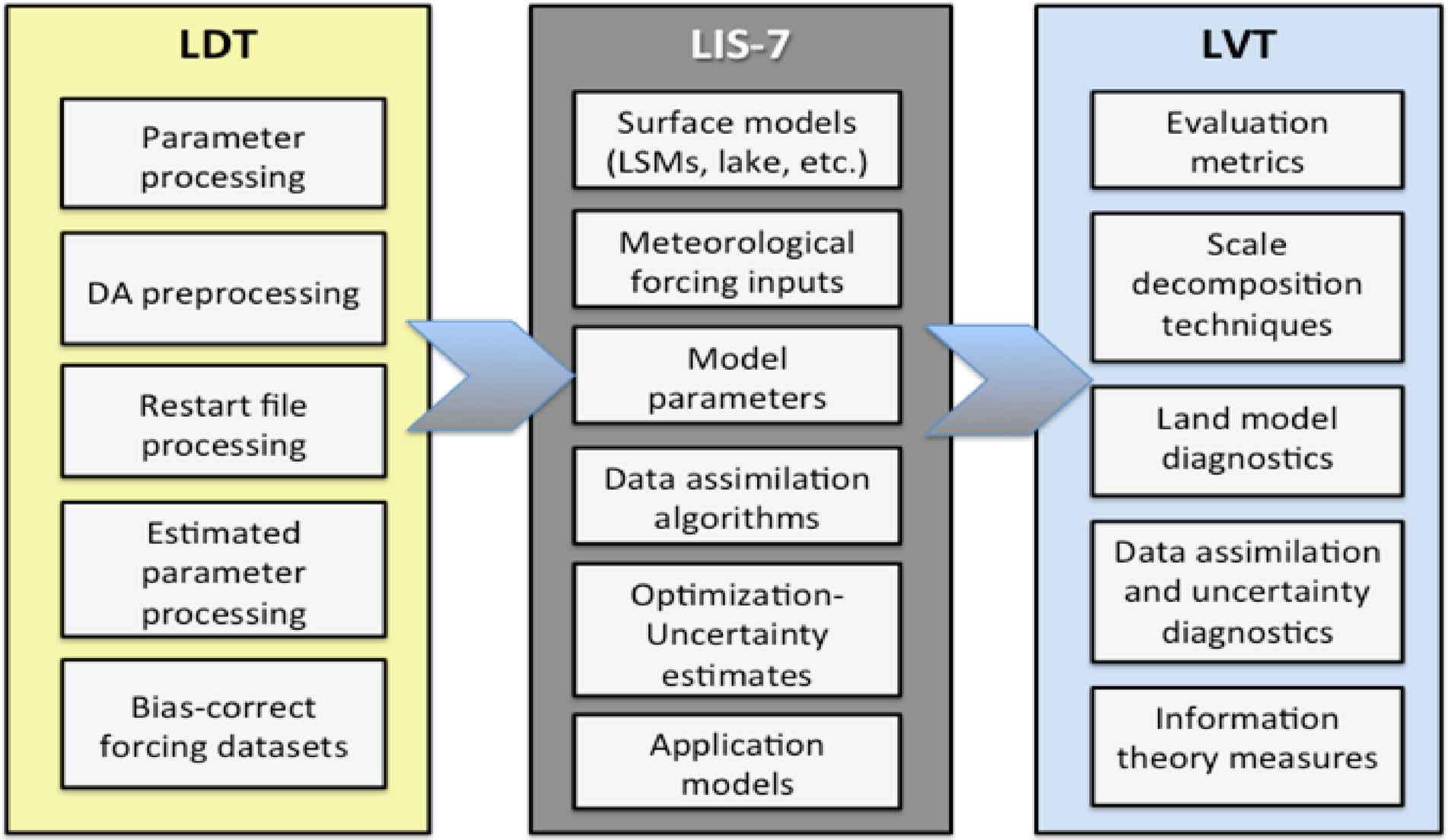
LDT's role in the LISF framework



The complete model-data fusion (MDF) paradigm for LDT-LIS-LVT.

LDT is the data processing environment that feeds into the modeling and data assimilation environment of LIS (and also LVT).

End-to-End Framework of LDT, LIS, LVT



LDT data format/input “philosophy”

- Model parameter and observational datasets should be read in from their *“native”* grids and file formats, with little to no preprocessing required.

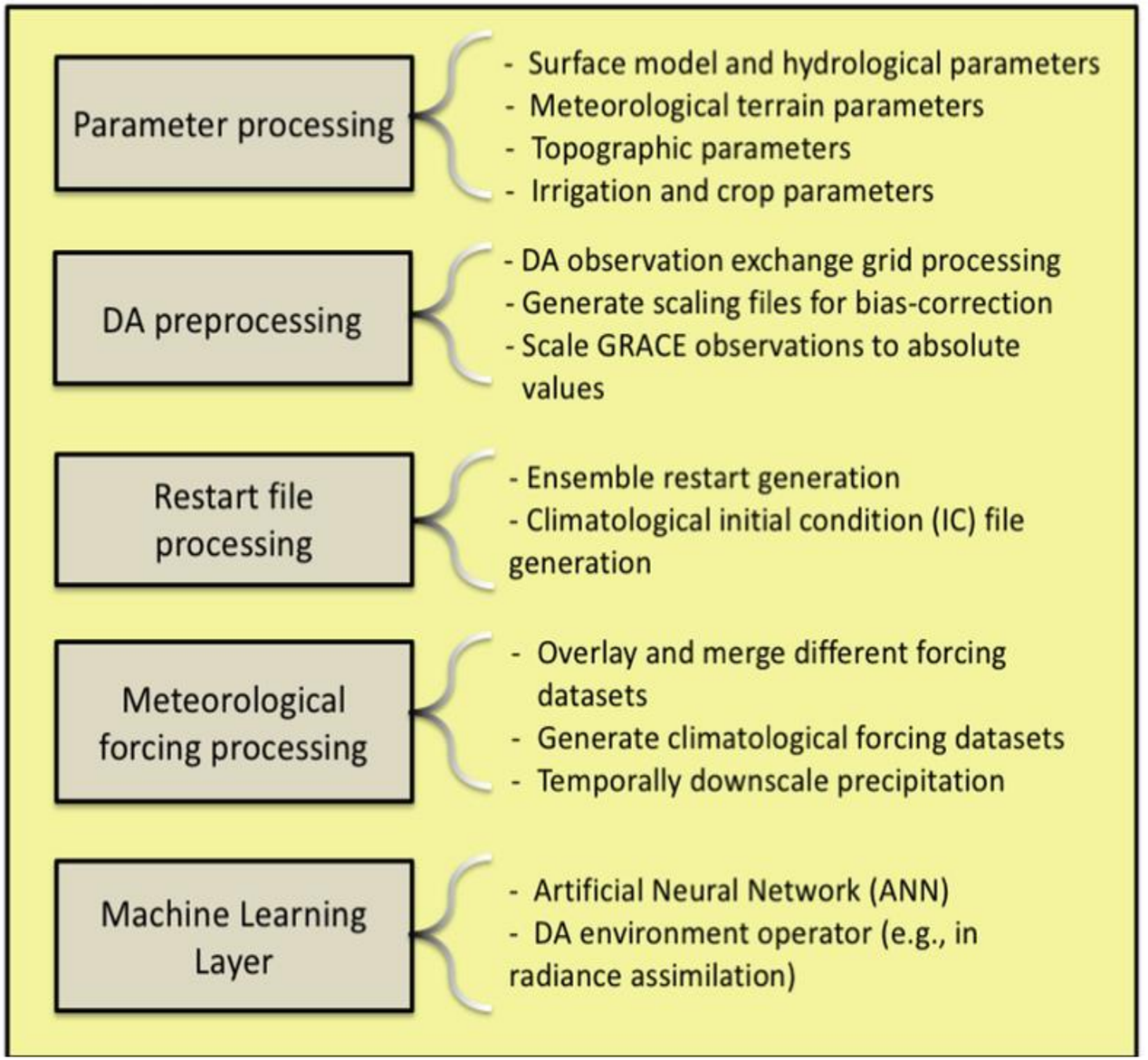
- LDT is designed to support many different data formats and files in their original form that are provided by a data center, government agency, etc.

- In the past, the LIS team had processed several different parameters onto a common binary format, which originally was provided to the community and had been typically referred to as “LIS-data”.

- We have moved away from this, as LIS is not a data center.

Current LDT Components

- **LDT supports a variety of different components and features that provide key inputs to LIS.**
- On the right, some of the main LDT components are highlighted and various features associated with each one.
- Today, we will present an overview of each component, but focus more on the **“Parameter processing”** component.



LDT "Run Mode" options

Option	Description
LSM parameter processing	Processes parameters required for a given model run within LIS and outputs to a common netcdf file
DA preprocessing	Prepares certain bias correction statistics and observation processing for data assimilation (DA) runs in LIS
Ensemble restart processing	Generates ensemble restart files, e.g., netcdf-4 file with header info
Metforce processing	Can process meteorological forcing files, like in LIS-7
Metforce temporal downscaling	Temporally downscale coarser time-step forcings with finer scale ones; works mostly with precip. for now
Statistical downscaling of met forcing ("climatology forcing")	Generate sub-daily, meteorological climatologies for a variety different forcing datasets.
ANN processing	Artificial neural networks and machine learning approaches are currently being implemented and tested in LDT

LDT parameter preprocessing

- Offers subsetting, reprojecting, aggregation, or downscaling and new tiling options for parameter files onto the targeted LIS grid and domain;
- Groups parameter files together into a common NetCDF-4 format file, which includes header and grid information for the different data layers;
- The input NetCDF-formatted file can be easily loaded and read in at LIS run-time;
- LDT also provides many run-time checks (e.g., checks that correct parameters were selected for a given model) and parameter statistics (e.g., land/water mask gridcell checks).

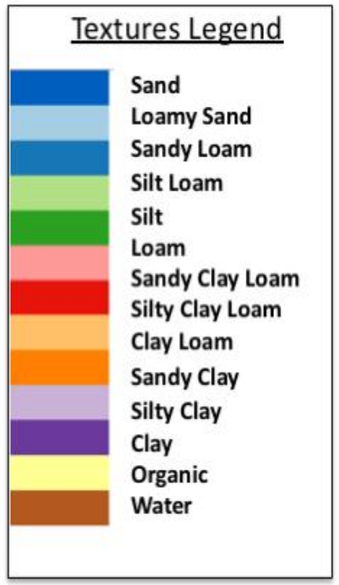
Types of 2D parameters (examples)

Parameter Type	Information	Description
Land cover	*Dominant vegetation type per gridcell, static*	Landcover maps typically come in a 2-D map at moderately high resolutions, e.g., ~ 1 km
Greenness fraction	*"gfrac," Monthly climatologies*	The fraction of vegetation greenness and typically varies by month and model-dependent
Albedo	*Monthly climatologies*	Monthly albedo maps, which typically reflect snow-free conditions or diffuse/direct components
Elevation	*Average elevation for gridcell*	Elevation and related topographic 2D fields typically at high resolutions (e.g., 90 meters)
HyMAP parameters	*Static maps that fit pre-processed domains*	HyMAP v1, v2 – Parameters on specific grids and resolutions for generating final input file for LIS

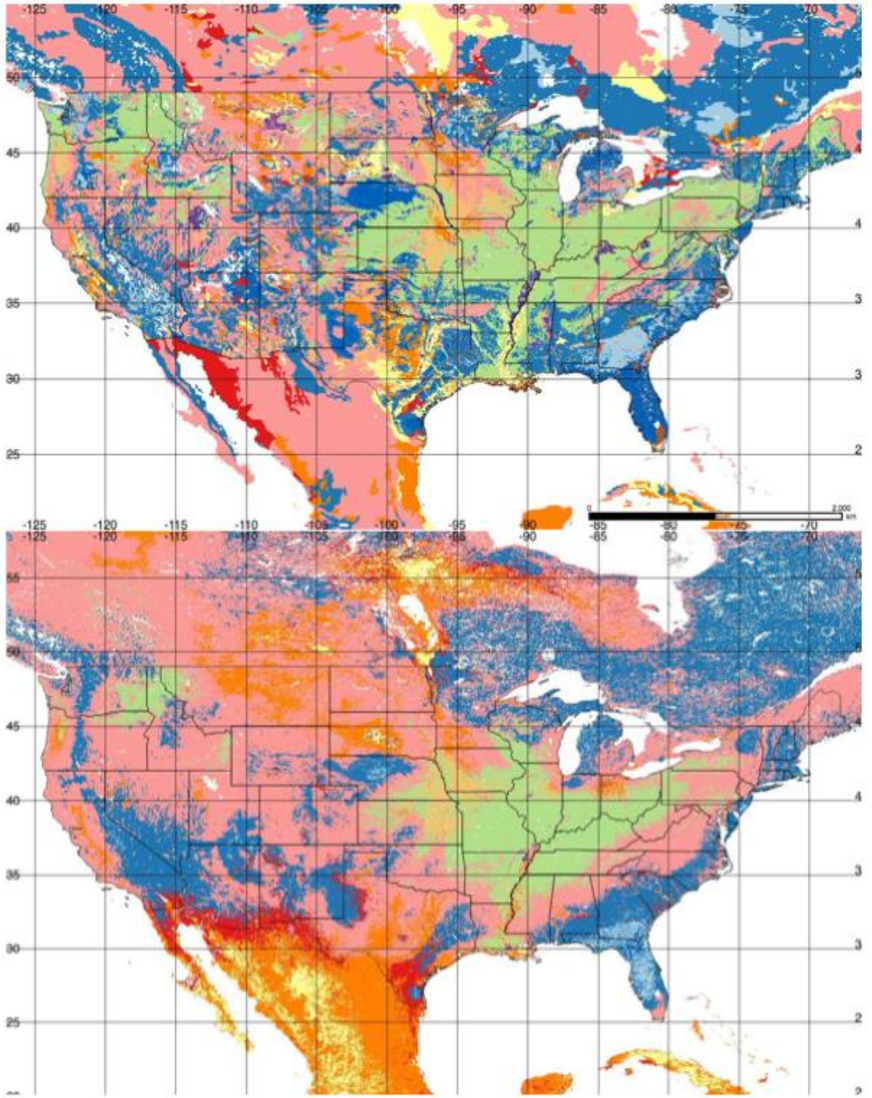
Example: Different soil texture class maps

- LDT supports a variety of different physiographic land parameter maps, for different LSMs, hydrology models, and other features (e.g., irrigation, crops, etc.).
- On the right is an example of different soil texture map sources supported within LDT, and which can be read into LIS to support land surface models requiring this parameter input.
- Both discrete and continuous type data sources are supported within LDT.

STATSGO-v1/FAO soil textures (1 km)



ISRIC soil textures (1 km)



LDT Grid, Projection, and other features

- Supports different projections, such as Plate-Carree (“lat-lon”), Lambert, and polar-stereographic.
- A variety of interpolation and upscaling routines are available for performing grid spatial transforms for input to LIS.
- Supports a variety of LSM and hydrological parameters and file formats.
- Generates run-time diagnostic information to a log file, as similarly done in LIS and LVT. Provides detailed diagnostic information, from warning to error messages to alert the user where the run-time failed.

LDT Spatial Transform Options

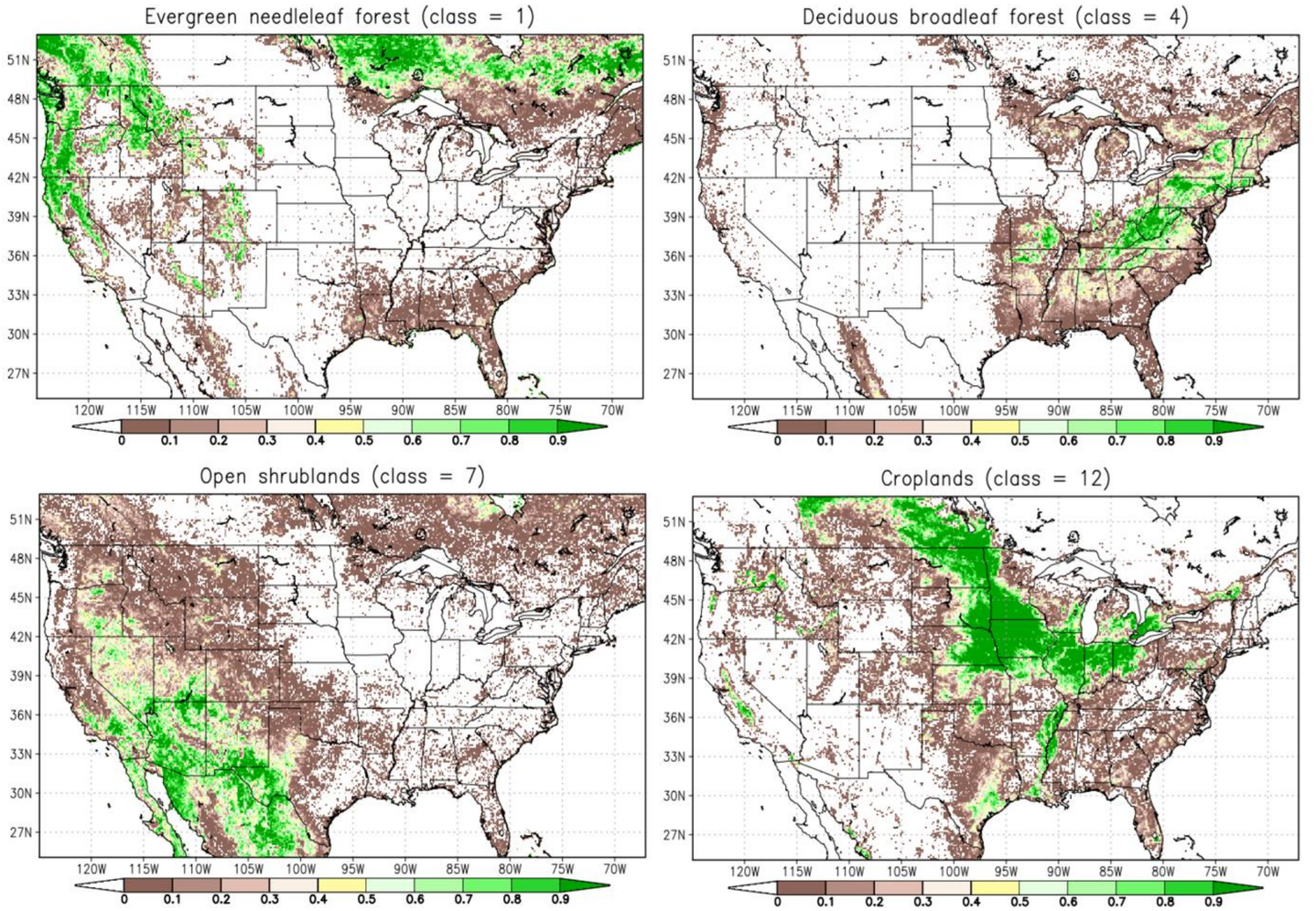
LDT Option	Entry Options	Description
[Parameter] spatial transform:	**none**	No scaling or transformation needed.
	average	Average finer resolution grid to coarser scale one (*upscale*).
	mode	Select dominant (mode) of finer scale gridcells to coarser one (*upscale*).
	tile	Apply accounting system for estimating coarser sub-grid "tiles" for output grid (*upscale*).
	neighbor	Apply nearest neighbor method to input grid (*upscale/downscale*).
	bilinear	Apply bilinear interpolation method to input grid (*downscale*).
	budget-bilinear	Apply conservative "budget" bilinear interpolation method to input grid (*downscale*).

Upscale options

Downscale options

Example: Fractional landcover classes

- Higher spatial resolution land cover (e.g., ~1 km) and other parameter datasets, such as soil texture, can be aggregated to coarser grid sizes using “tile” option.
- Each discrete class can be accounted for as a frequency or percentage of the total of vegetation class pixels occurring within each coarser gridcell.
- The example on the right shows the frequencies of four MODIS-based IGBP land classes, aggregated to 0.125-deg spatial resolution via tiling.
- This can also be done for soil texture and topographic maps, like elevation.



LDT Land cover processing options

LDT Option	Option	Description
Land cover classification:	*UMD, IGBPNCEP, USGS, and others*	Enter the land cover/land use classification scheme, which varies with different sources
Create or readin landmask:	*create*	Creates a landmask from the landcover file
	readin	Reads in a landmask file
Landcover map projection:	*latlon, and others (depending on the parameter read in)*	Projection/grid type for the input land cover parameter file.
Landcover spatial transform:	*e.g., tile, mode, nearest neighbor*	Mode for transforming the input parameter projection/grid to the LIS output (run-time) grid

Example: LDT Land cover configuration

"Native" or raw data entries:

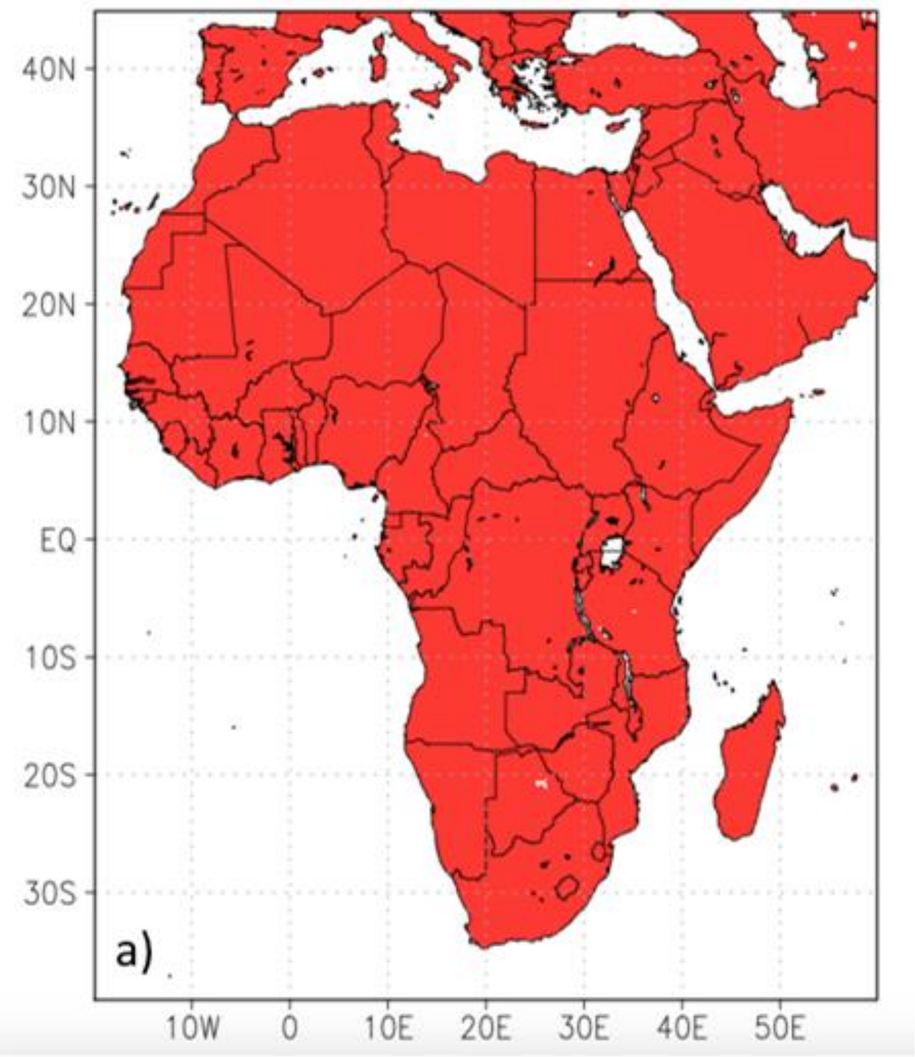
```
# Landcover/Mask Parameter Inputs
Landcover classification:       "IGBPNCEP"          # Enter land cover classification type
Landcover file:               ./input/igbp.bin      # Landcover map path
Landcover spatial transform:      tile              # none | mode | neighbor | tile
## Note: The fill entries here will 'fill' a missing landcover value with a neighboring one or assigned value.
Landcover fill option:          neighbor            # none | neighbor (Not needed if creating landmask)
Landcover fill radius:           3.                 # Number of pixels to search for neighbor
Landcover fill value:            5.                 # Static value to fill where missing
Landcover map projection:       latlon              # Landcover map projection
```

* Options for ensuring agreement between landmask and landcover (and other parameters).

LDT supports a variety of "native" format landcover and also landmask datasets. Above, we read in the MODIS-based, IGBP landcover dataset ("igbp.bin" = binary-format).

We would use the "neighbor" fill option, if we were to read in a landmask file that has a *different* land-water mask than our landcover map. This "fill" option ensures agreement between the landcover and landmask map files. This "imposes" the landmask file.

Importance of “masks” in LDT and LIS



- In LIS, the surface types (*such as land-surface points*) control which grid points are run during the model simulation.
- Because of this, each land point needs to have valid parameter values that the model will need to run at that grid point.
- If not, the model will crash or will produce undesirable results.
- Thus, we must *ensure agreement between the landmask and each model grid cell’s parameters*.

Example: LDT landmask config options

Read in Landmask entries:

```
# Create landmask field from readin landcover map or read in separate landmask file
Create or readin landmask:      "readin"                # create | readin
Landmask file:            ./input/landmask_UMD.1gd4r  # Land mask file (if needed to be read-in)
Landmask spatial transform:       mode                # none | mode | neighbor
## Note: The 'mode' entry here will determine the dominant mask value for each gridcell. ##
Landmask map projection:          latlon              # Landmask map projection
Landmask lower left lat:         -59.995              # Extents of the 1KM UMD landmask
Landmask lower left lon:        -179.995
Landmask upper right lat:         89.995
Landmask upper right lon:        179.995
Landmask resolution (dx):          0.01
Landmask resolution (dy):          0.01
```

"Creating" Landmask from Landcover entry:

```
# Create landmask field from readin landcover map or read in separate landmask file
Create or readin landmask:    "create"          # create | readin
Landmask file:                 none             # Land mask file (if needed to be read-in)
Landmask spatial transform:    none             # none | mode | neighbor
Landmask map projection:      latlon
```

Difference between LANDMASK & DOMAINMASK

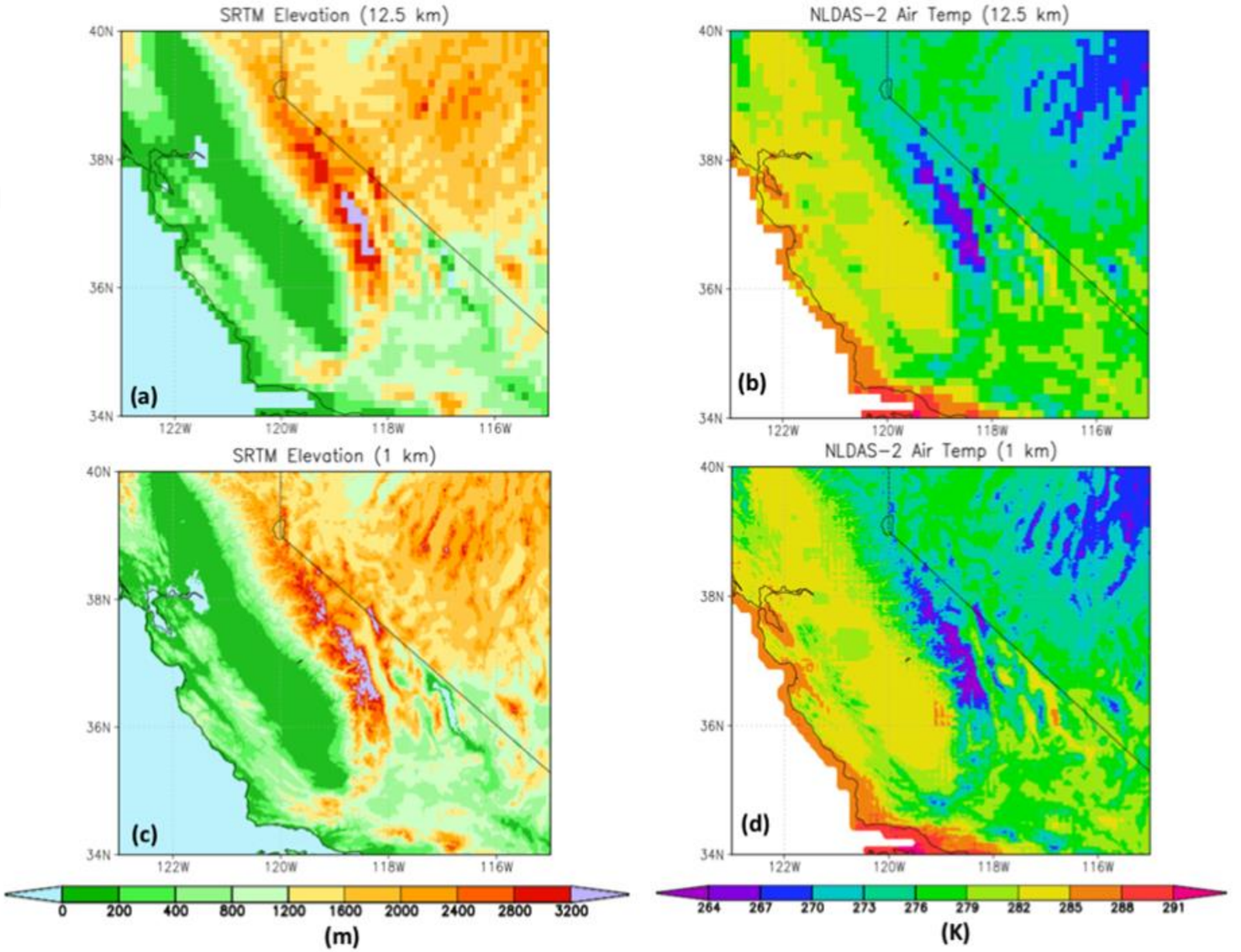
- The “**LANDMASK**”: The valid “land”-surface model grid point on which the LSM and the routing models will run
- The “**DOMAINMASK**”: The valid surface type grid points in the given domain, which can include “open water” and “lake” surface types
- These two “mask” files are important to keep track of when ensuring consistency between the model parameters, and the masks, and how LIS distinguishes between them.

Example: Types of 2-D parameters

Parameter Type	Information	Description
Land cover	*Dominant vegetation type per gridcell, static*	Landcover maps typically come in a 2-D map at moderately high resolutions, e.g., ~ 1 km
Greenness fraction	*"afrac," Monthly climatologies*	The fraction of vegetation greenness and typically varies by month and model-dependent
Albedo	*Monthly climatologies*	Monthly albedo maps, which typically reflect snow-free conditions or diffuse/direct components
Elevation	*Average elevation for gridcell*	Elevation and related topographic 2D fields typically at high resolutions (e.g., 90 meters)
HyMAP parameters	*Static maps that fit pre-processed domains*	HyMAP v1, v2 – Parameters on specific grids and resolutions for generating final input file for LIS

Example: “Downscale” meteorological data

- Using 30-arcsec SRTM (~1-km) elevation data to topographically “downscale” (or adjust) coarser-scale meteorological forcing fields, e.g., 2-m air temperature and surface pressure, using the environmental lapse-rate (ELR = 6.5 K km^{-1}).
- Upper-right: NLDAS-2 air temperature at native resolution of 12.5-km
- Lower-right: NLDAS-2 air temperature at downscaled resolution of 1-km
- Downscaling in LIS during runtime
- We are adding dynamic lapse-rate schemes into LIS.



LDT Data Assimilation observation processing

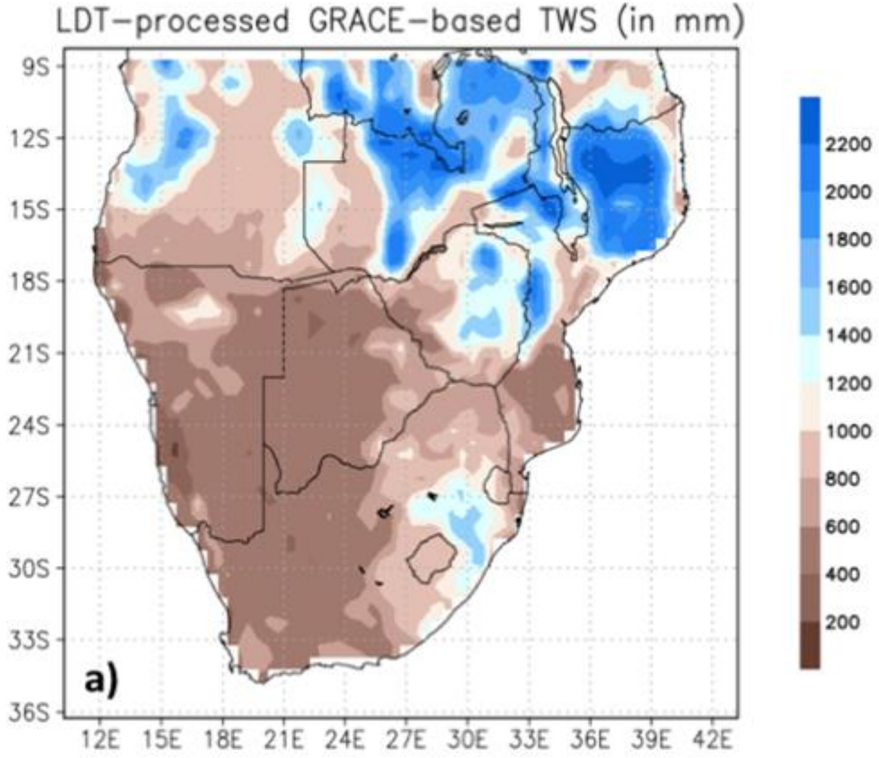
- Several remotely-sensed datasets are supported in LDT for pre-processing for data assimilation (DA) in LIS.
- LDT provides the DA observations - grid exchange processing (e.g., either in model- or observation-space).
- Generates the cumulative density function (CDF) files for normal-deviate based scaling, which performed in LIS for soil moisture bias correction.
- GRACE terrestrial water storage (TWS) anomaly datasets can be modified and have different types of scaling applied. Also surface water storage (SWS) can now be accounted for (using a routing model, like HyMAP).

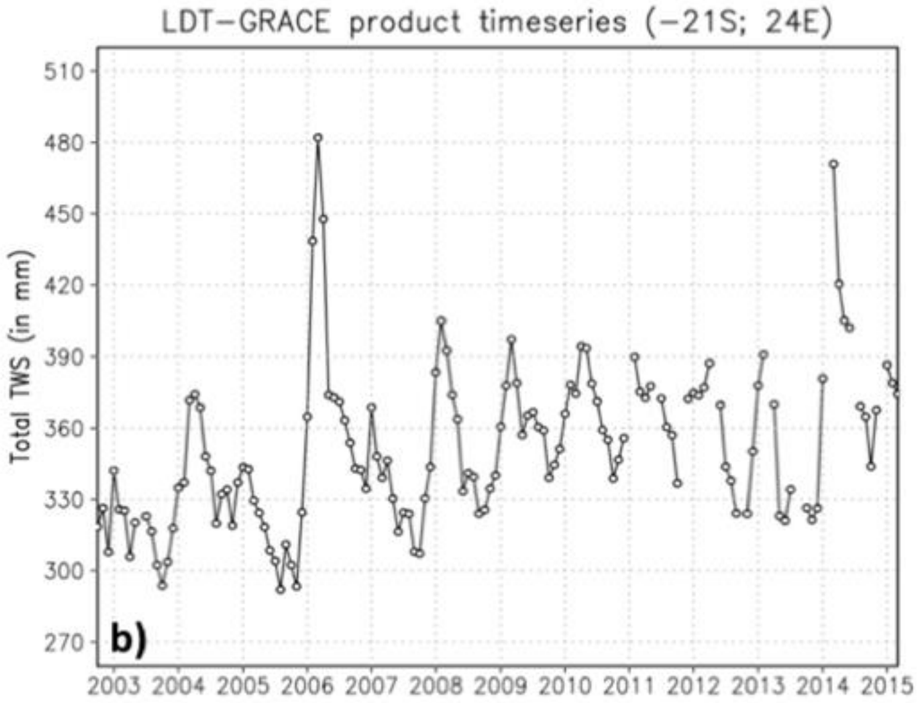
Example: LDT processing of GRACE-based TWS

LDT processes GRACE-based TWS anomalies by combining them with a long-term LIS-LSM retrospective or open-loop (OL) run to generate model-space TWS estimates to be assimilated during LIS runtime.

a) LDT-processed GRACE TWS (in mm) for Southern Africa domain.

b) Timeseries of LDT-generated GRACE TWS (in mm) at -21.0S and 24.0E.





LDT Initial Condition processing options

- Generate an ensemble restart file from a single deterministic restart file to initialize a DA or seasonal forecast run.
- Generate an ensemble mean restart file from an ensemble-based restart file.
- Can generate a climatological restart file from a 30-year record (given the same date for those 30-years).
- File format generated: netcdf-4.
- This feature will be covered further in the LIS data assimilation part of our tutorial.

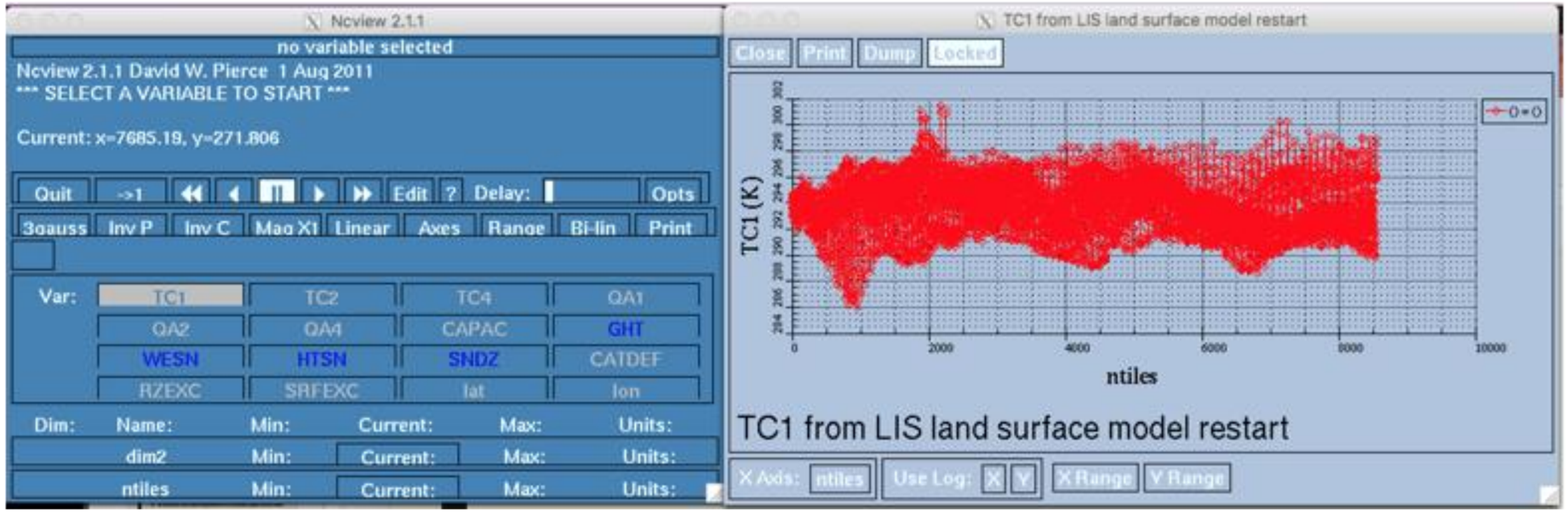
Example: LDT initial condition processing

Number of tiles of the OL restart file

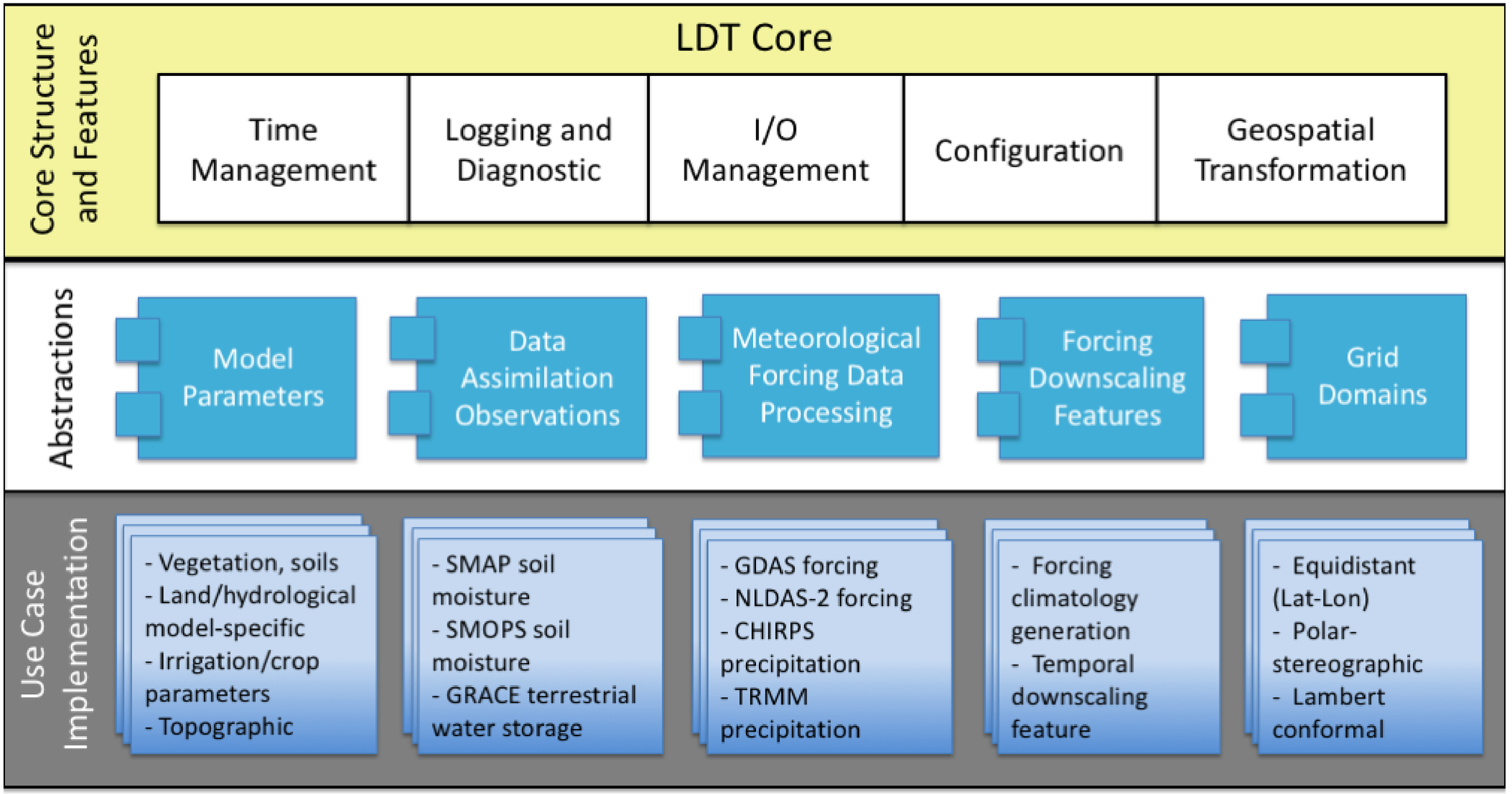
```
ncdump -h LIS_RST_CLSMF25_201512312345.d01.nc
  netcdf LIS_RST_CLSMF25_201512312345.d01 {
  dimensions:
     ntiles = 8552 ;
```

Number of tiles of *4-member* restart file

```
ncdump -h clsm_ol_ensrst_201512312345.nc
  netcdf clsm_ol_ensrst_201512312345 {
  dimensions:
     ntiles = 34208 ;
```



LDT structure



Different data formats in LDT

Format Type	Example Reader (*parms* directory)	Data Type
Text-based (ASCII)	albedo/read_RobinsonKuklaNative_mxsnoalb.F90	Continuous
Binary	albedo/read_RobinsonKuklaNative_mxsnoalb.F90	Continuous
NetCDF	mask/read_maskfile_UKMO_CAP_Netcdf.F90	Discrete
Tiff / GeoTiff	soils/read_ISRIC_texture.F90	Discrete
HDF5	albedo/read_BarlageNative_maxsnoalb.F90	Continuous
BIL	GeoWRSI/read_GeoWRSI2_lgp.F90	Continuous

LDT: Journal article and documentation

For the latest LDT reference with a more complete description of the given components and features, please see:

Arsenault, K.R., S.V. Kumar, J.V. Geiger, S. Wang, E. Kemp, D.M. Mocko, H.K. Beaudoing, A. Getirana, M. Navari, B. Li, J. Jacob, J. Wegiel, and C.D. Peters-Lidard: The Land surface Data Toolkit (LDT v7.2) – a data fusion environment for land data assimilation systems, Geosci. Model Dev., 11, 3605-3621, https://doi.org/10.5194/gmd-11-3605-2018, 2018.

https://www.geosci-model-dev.net/11/3605/2018/gmd-11-3605-2018.html

Also, visit our LDT webpage (located within the LIS website):

https://lis.gsfc.nasa.gov/ldt-documentation

https://lis.gsfc.nasa.gov/ldt-data